

PROJECT TITLE

AD4V71 - Machine Learning Operations (MLOps)

UPI SHIELD

Real Time UPI Transaction Anomaly Detection with MLOps Pipeline

Presenting By

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Problem Identification and Title Justification

Relevance of the problem

- UPI processed 24,000+ crore transactions in FY 2025-26 — India's dominant digital payment rail
- Static, rule-based fraud detection can't scale to this volume or adapt to evolving fraud patterns
- Real-time, learning-based anomaly detection is needed to keep pace with fraud at this scale

Clarity of problem statement

Need: a real-time, intelligent system to detect anomalous UPI transactions — one that learns and adapts, not just follows fixed rules
Solution: UPI Shield — a complete MLOps pipeline (training, deployment, monitoring), not just a one-off model

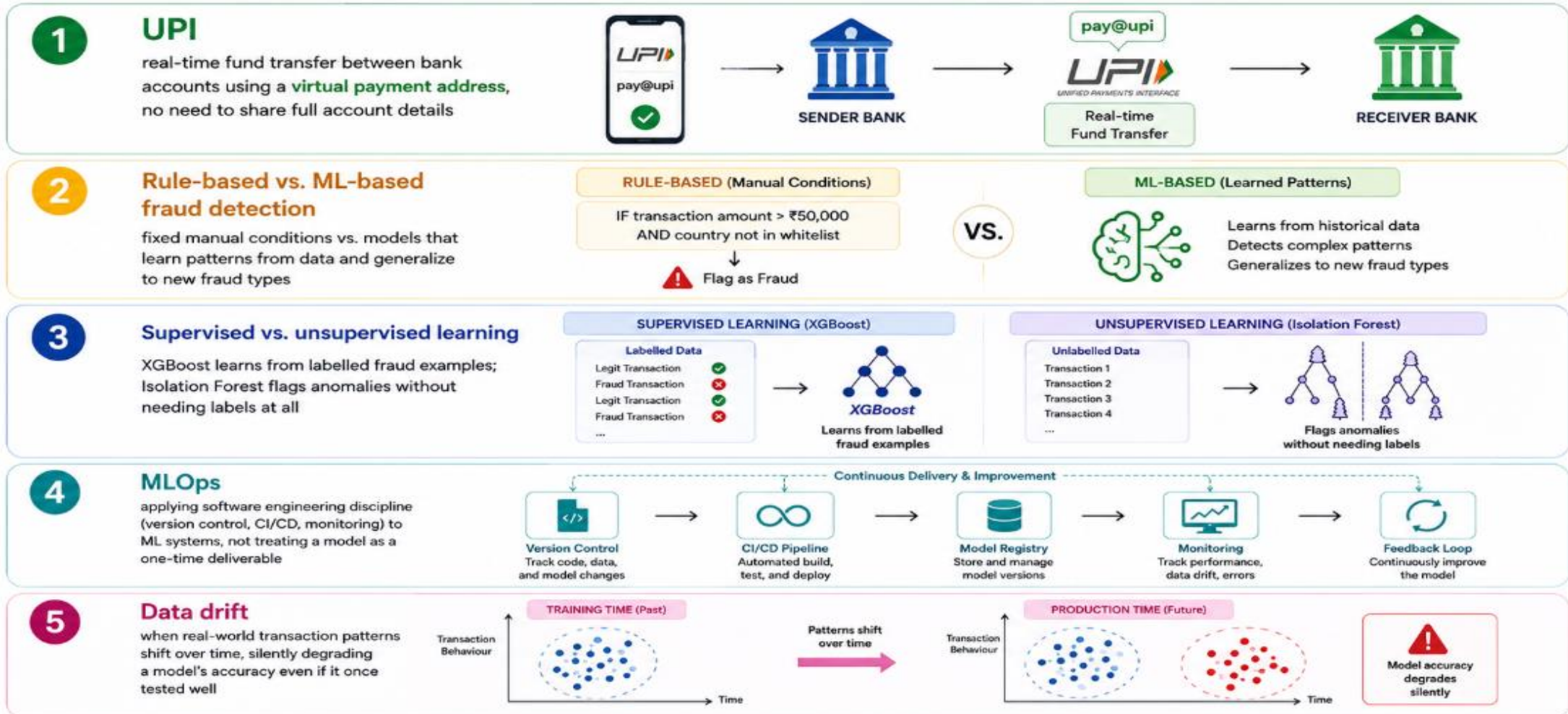
Justification of project title and scope of the project

"UPI" anchors the project to a specific, high-stakes domain — not generic payment fraud "Shield" reflects its function: a protective layer that flags anomalies before they cause harm

Scope: EDA → modelling → API → dashboard → drift monitoring, trained on Pay Sim data with real NPCI statistics as a drift benchmark; no real user data is processed

Problem Identification and Title Justification

Basic concepts related to the project



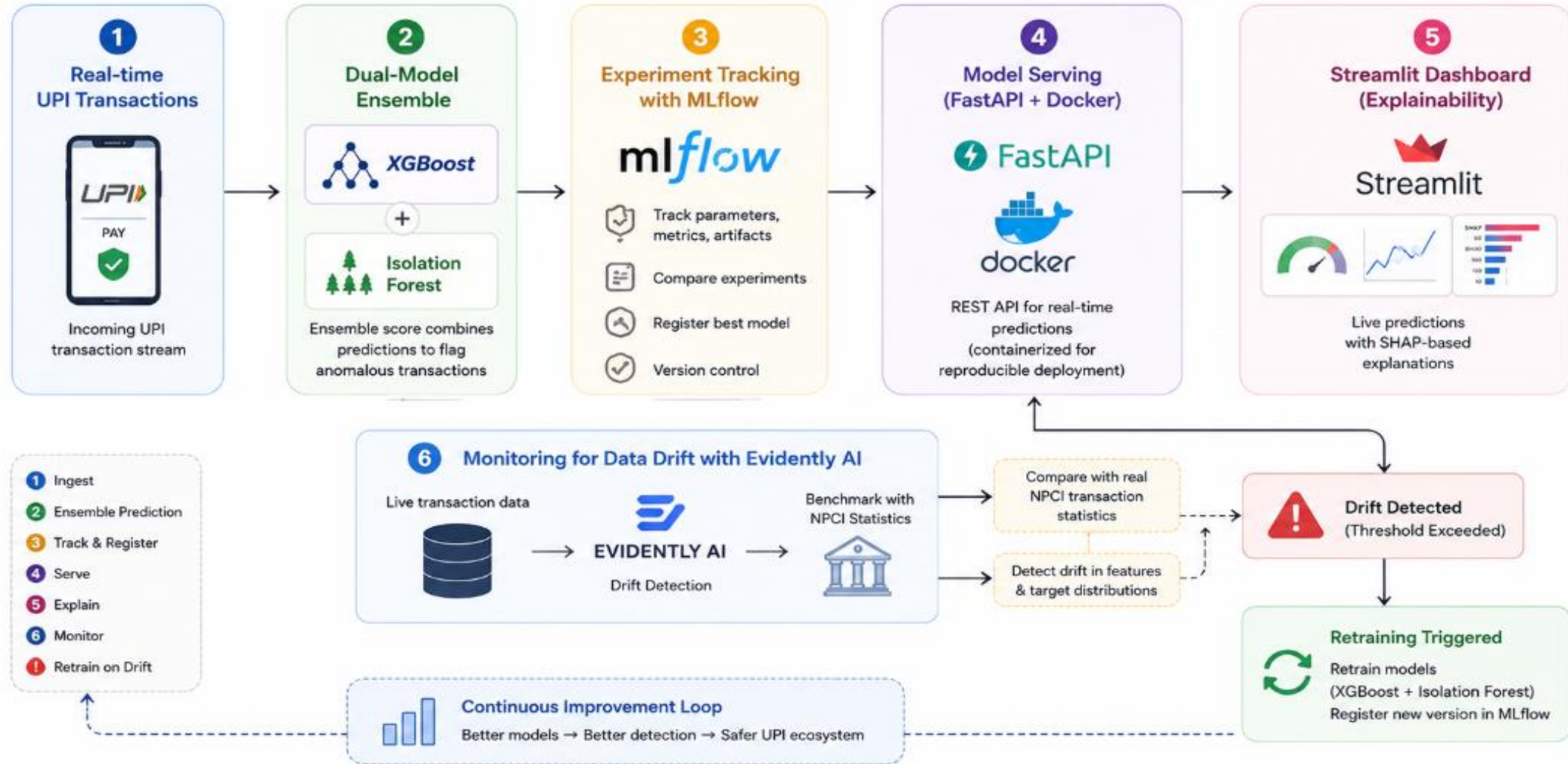
Literature Survey

- **1. Meena, Harshitha & Kavitha (2025)** — compared Random Forest, Isolation Forest, One-Class SVM for UPI fraud
Advantage: rigorous comparison across model types with ROC-curve evaluation
Disadvantage: doesn't address real-time deployment or drift monitoring
- **2. Rani et al. (2024)** — XGBoost + SMOTE + PCA, proposed biometric authentication layer
Advantage: directly handles class imbalance and reduces dimensionality
Disadvantage: PCA reduces feature interpretability; biometric layer adds deployment complexity
- **3. Systematic Review, UPI fraud 2016-2025** — surveyed CNN/LSTM approaches, fraud vectors (phishing, QR-code, KYC fraud)
Advantage: broad view of the full fraud threat landscape
Disadvantage: a review only — no working system implemented or tested
- **4. Kamble et al. (2025)** — stacked-generalization ensemble for UPI fraud
Advantage: ensemble improves robustness over single-model approaches
Disadvantage: heavier compute cost, harder to run in real-time at low latency
- **5. IEEE Hybrid DQN-XGBoost (2025)** — reinforcement learning + gradient boosting, 98.7% accuracy
Advantage: high accuracy, low false-positive rate (1.2%), adaptive decision-making
Disadvantage: RL models are complex to train/reproduce; tested on generic Kaggle data, not UPI-specific

Research gaps identified: none of these combine a supervised + unsupervised ensemble the way UPI Shield does, none benchmark against real NPCI drift data, and most stop at model accuracy without addressing deployment or monitoring.

Related work, in one line: existing UPI fraud research is strong on model accuracy but weak on production-readiness — which is exactly the gap UPI Shield is built to fill.

Objectives of the Project



Project Planning and Time Schedule

Gantt chart / timeline preparation



Project Planning And Timing Schedule

Task allocation among team members

Mohana Priya (Data & Modelling Lead) — EDA, feature engineering, XGBoost + Isolation Forest ensemble, MLflow tracking, SHAP explainability

Priyanka (MLOps & Deployment Lead) — FastAPI service, Docker, Streamlit dashboard, CI/CD pipeline, Evidently drift monitoring, cloud deployment

Milestone planning

M1: EDA complete — End of Week 1

M2: Best model registered in MLflow — End of Week 3

M3: API + dashboard live and deployed — End of Week 4

M4: Drift monitoring operational— End of Week 5

M5: Final polish complete— End of Week 6

References

- Lopez-Rojas, E. A., Elmir, A., & Axelsson, S. (2016). PaySim: A financial mobile money simulator for fraud detection. *The 28th European Modeling and Simulation Symposium (EMSS)*, Larnaca, Cyprus.
- Meena, V., Harshitha, M., & Kavitha, S. (2025). Enhanced UPI Fraud Detection Using Machine Learning: A Proactive Approach to Secure Digital Transactions. *International Journal for Research Trends and Innovation (IJRTI)*.
- Rani, et al. (2024). UPI fraud detection using XGBoost, SMOTE, and PCA. As referenced in: *Mobile payment fraud detection in UPIs through machine learning techniques: A systematic review* (2025).
- Kamble, V. B., Pisal, K., Vaidya, P., & Gaikwad, S. (2025). Enhancing UPI Fraud Detection: A Machine Learning Approach Using Stacked Generalization. *International Journal of Multidisciplinary on Science and Management*, 2(1), 69-83.
- Enhancing Digital Payment Security: UPI Fraud Detection with Advanced Machine Learning Algorithms. (2025). *IEEE Conference Publication*.
- PaySim dataset: kaggle.com/datasets/ealaxi/paysim1
- National Payments Corporation of India — UPI transaction statistics: npci.org.in

Thank You!